

Continuous glucose monitoring for glycemic control in non-insulin-treated patients with type 2 diabetes

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ABSTRACT

Background: Continuous glucose monitoring (CGM) has been associated with improved glycemic control in individuals with non-insulin-treated type 2 diabetes (T2D), but adoption in endocrinology clinics remains limited.

Local Problem: This quality improvement (QI) project aimed to explore the use of CGM for non-insulin-treated patients with T2D within an endocrinology clinic in Los Angeles County and to evaluate feasibility of its adoption in this practice setting.

Methods: It was a retrospective chart review.

Interventions: Eligible patients were adults (≥ 18 years) with T2D and hemoglobin A1c (HbA1c) $> 7\%$ who had not been treated with insulin. Ten patients met inclusion criteria. Descriptive statistics summarized HbA1c and time-in-range (TIR) over 6 months. An interrupted time series was also conducted on two patients with sufficient longitudinal data to assess HbA1c changes before and after CGM use.

Results: After 6 months of initial CGM use, 8 of 9 patients achieved HbA1c reductions of $\geq 0.3\%$. One patient demonstrated a $\geq 8\%$ increase in TIR with adequate device use. Interrupted time series analyses illustrated individual HbA1c trajectories, showing immediate reductions after CGM initiation and nonsignificant downward trends over time.

Conclusion: This QI project showed that patients with T2D treated with noninsulin medications successfully initiated and used CGM. Early improvements in HbA1c were observed; however, sustained glycemic outcomes varied depending on consistent CGM wear and adherence to follow-up, underscoring the need for individualized support. Nurse practitioners can play a key role in promoting sustained CGM use through structured diabetes education, integration of CGM data into lifestyle counseling, and regular follow-up.

Keywords: CGM; Continuous glucose monitoring; non-insulin-treated type 2 diabetes; T2D.

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Introduction

Diabetes is a progressive, chronic metabolic disorder that affects approximately 11.6% of the US population (Centers for Disease Control and Prevention [CDC], 2024). Uncontrolled diabetes can lead to significant microvascular and macrovascular complications, including diabetic neuropathy, nephropathy, retinopathy, coronary artery

disease, peripheral artery disease, and cerebrovascular disease. These complications increase the risk of premature mortality (Yamazaki et al., 2018), with diabetes ranking among the top 10 causes of mortality in North America in 2021 (CDC, 2024). The economic burden of diabetes in the United States is substantial as well, with an estimated cost of \$413 billion in 2022, comprising \$307 billion in direct costs and \$106 billion in indirect costs (CDC, 2024).

Type 2 diabetes mellitus (T2D) accounts for 90%–95% of all diabetes cases (CDC, 2024). Management typically begins with noninsulin pharmacotherapy, lifestyle interventions, and consideration of cardio-renal risk factors. However, as the disease progresses, insulin therapy may become necessary (American Diabetes Association [ADA], 2024b). Despite its benefits, insulin initiation presents challenges such as patient adherence, motivation, fear of hypoglycemia,

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socioeconomic factors, and treatment complexity among barriers to effective use. (Alidrisi et al., 2021).

Self-monitoring of blood glucose using conventional glucometers remains a widely used approach for glycemic management in non-insulin-treated T2D. However, its effectiveness remains debated, with limited research supporting its impact on glycemic control (Madeo et al., 2020). Challenges associated with glucometer use include discomfort, pain, bleeding, and the complexity of sample collection and disposal (Chen et al., 2018; Price & Walker, 2016).

Available knowledge

Continuous glucose monitoring (CGM) is an advanced alternative for tracking blood glucose levels. By measuring interstitial glucose, CGM systems provide several advantages over traditional glucometers, including improved usability, greater accuracy, trend monitoring, dietary tracking, and alerts for both hypoglycemia and hyperglycemia, making them a superior option for many patients (ADA, 2024b). Evidence shows that CGM is effective in supporting glycemic management for both type 1 diabetes and T2D. Continuous glucose monitoring serves not only as a monitoring tool, but it also serves as a motivational device that encourages lifestyle modification and enhances self-care, particularly among individuals using insulin therapies (Beck et al., 2017; Carlson et al., 2017; Jackson et al., 2021; Martens et al., 2021). More recently, studies have extended these findings to adults with T2D who are not using insulin. Across diverse settings, CGM use has been associated with improved glycemic control, increased time in range (TIR), reduced hypoglycemia, and greater patient satisfaction when paired with interventions such as increased frequency of use (Bergenstal et al., 2022; Moon et al., 2023), lifestyle modifications (Cox et al., 2020; Moon et al., 2023), diabetes education (Aronson et al., 2023; Cox et al., 2020), or continuous surveillance (Price et al., 2021). Collectively, this evidence underscores the clinical value of CGM as both a monitoring and motivational tool, supporting its integration into routine diabetes care to improve outcomes in this population.

Local Problem

Although CGM is not yet considered the standard of care for patients with T2D managed without insulin, Level I evidence indicates that its use can lower HbA1c by approximately 0.31% and increase TIR by about 8.63% (Ferreira et al., 2024). Nevertheless, adoption remains low in real-world practice, with only about 5% of eligible patients in endocrinology settings currently using CGM (Mayberry et al., 2023). Despite readily available technology and services, a similar underutilization of CGM exists in the endocrinology clinic where this project was conducted. This discrepancy represents a missed opportunity to optimize glycemic outcomes, improve patient engagement, and reduce disparities in diabetes care.

Rationale/conceptual framework

The Donabedian model (1966/2005) served as the guiding framework for this project. The Donabedian model conceptualizes health care quality through structure, process, and outcome measures. Structure reflects the physical and organizational context of care, process reflects the delivery of services and treatments, and outcome reflects the effects of care on patient health (ACT Academy, n.d.; Donabedian, 1966/2005). This framework provided a structured approach to examine how real-time CGM was supported, implemented, and linked to patient outcomes in the clinic.

Specific aim

This project aimed to explore the use of CGM on glycemic outcomes in non-insulin-treated patients with T2D within an endocrinology clinic and to identify opportunities for improving workflows and nurse practitioner (NP)-lead support in clinical practice.

Methods

Setting and participants

The setting of this project was an ambulatory endocrinology clinic in Los Angeles County where CGM technology was used to improve glycemic control in patients with poorly controlled, non-insulin-treated T2D. The team included the principal investigator, a Doctor of Nursing Practice candidate at the time of the study, an endocrinologist who served as the clinical mentor, and three Doctor of Philosophy (PhD)-prepared nurses who provided guidance during the duration of the project.

Our inclusion criteria included adult patients (aged 18 years and older) diagnosed with T2D using the International Classification of Diseases, 10th Revision (ICD-10), with an HbA1c level $\geq 7\%$ at baseline documented in the electronic health record (EHR). Patients were also required to be on at least one noninsulin antidiabetes medication. Noninsulin therapies included biguanides, sulfonylureas, sodium-glucose cotransporter-2 (SGLT2) inhibitors, glucagon-like peptide-1 (GLP-1) receptor agonists, glucose-dependent insulinotropic polypeptide/GLP-1 receptor agonists, dipeptidyl peptidase-4 (DPP-4) inhibitors, meglitinides, thiazolidinediones, and α -glucosidase inhibitors. Patients were excluded if they had type 1 diabetes mellitus, gestational diabetes, or documented noncompliance with their care plan as recorded in the EHR. In addition, patients who used insulin (either basal or bolus) after starting CGM during the project were excluded.

Design

We conducted a retrospective chart review of patient records from November 2023 to November 2024. The review focused on HbA1c levels and TIR before and after CGM implementation to assess short-term glycemic

outcomes in the clinic population. The report followed the Standards for quality improvement (QI) Reporting Excellence (SQUIRE 2.0) guidelines to ensure transparent reporting of context, design, measures, and outcomes (Ogrinc et al., 2015).

Measures

Consistent with the Donabedian framework, measures were organized into structure, process, and outcomes.

Structure. Structure was assessed by the availability of a digital infrastructure in the endocrinology clinic to support CGM data collection and integration with the EHR. This infrastructure included software and network systems that transmitted CGM data, either through direct cable connections or via cloud-based platforms operated by the CGM manufacturer, from patients' devices to the EHR. Within the EHR, clinic staff and providers accessed and reviewed the data in the Media section to inform and support ongoing diabetes management.

Process. Process measures included the frequency of HbA1c testing and the review of CGM metrics during each diabetes visits, the duration of follow-up intervals between visits, and CGM device activity levels (percentage of time worn).

Outcomes. Outcomes were evaluated by changes in glycemic control within 6 months of real-time CGM implementation. Specifically, outcomes included the proportion of patients who achieved 1) a reduction in HbA1c levels of at least 0.3% and 2) an increase in TIR of at least 8%. Evidence shows that CGM use can lower HbA1c by approximately 0.31% in non-insulin-treated patients with T2D. In addition, CGM use has been shown to increase TIR by about 8.63% in the target population (Ferreria et al., 2024).

Follow-up intervals. The number of days between visit 0 (CGM initiation), visit 1, and visit 2 was calculated to describe the real-world timing of follow-up in relation to ADA-recommended 3-month intervals.

HbA1c. HbA1c is the gold standard for long-term glycemic control, reflecting mean glucose levels over the preceding 3 months. The American Diabetes Association (ADA, 2024b) recommends <7% as the target for most nonpregnant adults with T2D. According to Samson et al. (2023), the American Association of Clinical Endocrinology (AACE) suggests $\leq 6.5\%$ when achievable without hypoglycemia.

Time in range. Time in range measures the percentage of time blood glucose remains between 70 and 180 mg/dL, providing insight into glucose variability and daily management. Time below 70 mg/dL indicates hypoglycemia and time above 180 mg/dL indicates hyperglycemia, both of which should be minimized (ADA, 2024a).

Continuous glucose monitoring activity. Continuous glucose monitoring activity was assessed as the percentage of time devices worn, with $\geq 70\%$ activity over 14 days

considered adequate data capture, consistent with ADA (2024b) and AACE (n.d.) recommendations.

Data collection

Data were collected through a manual chart review of patients who met the inclusion criteria, using secure remote access to the EHR system, which allowed the first author to review patient charts off site from November to December 2024. HbA1c levels were assessed at baseline (visit 0, CGM initiation), first follow-up (visit 1), and second follow-up (visit 2). Time in range data were only available at visits 1 and 2 because CGM was not in use before visit 0, and therefore no baseline TIR could be obtained. Continuous glucose monitoring data were extracted only when devices achieved $\geq 70\%$ wear time over 14 days, as recommended by the ADA (2024b) and the AACE (n.d.). For each patient, HbA1c changes were analyzed from visit 1 to 2 for each patient, and TIR changes from visit 1 to 2.

Data analysis

Descriptive statistics. Descriptive statistics (measures of central tendency, variability, and frequency distributions) were used to summarize patient demographics and outcomes. Data were calculated for the percentage of patients achieving an HbA1c reduction of $\geq 0.3\%$ from visit 0 to visit 2 and a TIR improvement of $\geq 8\%$ from visit 1 to visit 2, with $\geq 70\%$ CGM activity in the preceding 14 days. The mean was used for the follow-up intervals from visit 0 to visit 1, from visit 1 to visit 2, and from visit 0 to visit 2.

Interrupted time series analysis. Single-case, patient-level interrupted time series (ITS) analyses were performed for two patients with sufficient pre- and post-CGM data points due to the small sample. These single-case ITS models were designed to illustrate temporal changes in HbA1c for individual patients rather than to estimate population-level effects. Model-based 95% confidence intervals were reported for transparency. These intervals represent uncertainty around individual trajectories and are not intended as population-level inference. Segmented regression was used to model changes in HbA1c level and trend while adjusting for medication dosage: $Y_t = \beta_0 + \beta_1(\text{time}) + \beta_2(\text{level}) + \beta_3(\text{trend}) + \beta_4(\text{medication dosage}) + \epsilon$, where β_0 estimates the baseline HbA1c level at the start of the time series, β_1 the pre-CGM trend of HbA1c (in days), β_2 the immediate change in HbA1c post-CGM, β_3 the change in the slope (trend) of HbA1c post-CGM, and β_4 the effect of medication dosage on HbA1c. Autocorrelation was assessed using the Durbin-Watson statistic and corrected for detected autocorrelation through stepwise autoregression. Model fit was examined using residuals and autocorrelation functions. Statistical significance was set at an alpha level of 0.05. Analyses were conducted in R (version 2024.12.1).

Ethical considerations

The University of California-Los Angeles Institutional Review Board exempted this project and granted a waiver of informed consent. All identifiable data were deidentified, securely stored, and accessible only to authorized personnel, with the code key deleted after data extraction. The project followed all institutional data privacy and security guidelines.

Results

Sample characteristics

Ten patients were included. The mean age was 60.9 ± 15.21 years (median = 55, range = 47). Six patients were male (60%), three female (30%), and one nonbinary (10%). Three patients identified as Asian (30%), three as White (30%), and one as Hispanic or Latino (10%); two patients (20%) had missing ethnicity data.

Data availability

Ten patients met inclusion criteria, but analytic sample sizes varied across outcomes due to missing data. **Table 1** summarizes the number of patients contributing data for follow-up intervals, HbA1c, TIR, CGM activity, and the ITS analysis.

Follow-up intervals

The mean time from visit 0 to visit 1 was 110.7 days (Mdn = 108.5, range = 143), and from visit 1 to visit 2 was 120.1 days (Mdn = 114.5, range = 255). The total mean duration from visit 0 to visit 2 was 230.7 days (Mdn = 218.5, range = 373), with one outlier.

HbA1c outcomes

At visit 1, HbA1c data were available from eight patients. All achieved reductions of ≥0.3% from baseline (mean −1.9%, median = −1.6%, range −0.3% to −3.6%). At visit 2, HbA1c data were available from nine patients. Eight patients had statistically significant changes in their HbA1c, whereas one patient had minimal changes

(−0.1%). The overall mean HbA1c was −1.7%, (median = −1.5%, range −3.6% to −0.1%; **Figure 1** and **Table 2**).

Time in range outcomes

Time in range data were available for nine patients across visits, with TIR1 available for nine patients and TIR2 for all 10 patients. From visit 1 to visit 2, the overall mean change was −3.2% (median = 0%, range −32% to +9%). One patient improved by more than 9%. Three patients had modest improvements (mean = +2.3%, range +1–4%). One patient had no change, and four patients declined (mean = −11.3%; median = −6.0%, range −32% to −1%; **Figure 2** and **Table 2**).

Continuous glucose monitoring activity

At visit 1, CGM activity data were available for seven patients (mean = 64.1%, median = 78.0%, range 7–98%); three had <70% and four had ≥70% CGM activity levels. At visit 2, 7 patients again had complete data (mean = 73.7%, median = 81.9%, range 29–93%); 2 had <70% and 5 had ≥70% CGM activity levels. Only one patient achieved both ≥8% TIR improvement and ≥70% CGM activity (**Figure 2** and **Table 2**).

Interrupted time series results

Results from the patient-level ITS models are presented for two patients. These findings are descriptive and reflect individual trajectories, not population-level estimates.

Patient ID#1. The baseline HbA1c was 6.22% (95% confidence interval [CI]: 2.05–10.40, $p = .009$). The pre-CGM trend was minimal ($\beta_1 = 0.001$ per day, 95% CI: −0.001 to 0.003, $p = .279$). The immediate post-CGM change was −0.719 (95% CI: −4.79 to 3.35, $p = .733$). The post-CGM trend was −0.865 per day (95% CI: −1.76 to 0.028, $p = .075$). Medication dosage showed no significant association ($\beta_4 = 0.153$, 95% CI: −2.53 to 2.83, $p = .912$). Autocorrelation was addressed using a corARMA ($p = 2$) structure, which provided the best model fit (AIC = 99).

Table 1. Data availability by outcome and visit

Outcome	Visit 1 (n)	Visit 2 (n)	Notes
Follow-up intervals	8	9	Two patients were missing follow-up data for visit 1, and one patient was missing follow-up data for visit 2
HbA1c	8	9	Missing data at visit 1 for 2 patients
TIR (visit 1 → visit 2 change)	9	10	One patient lacked visit 1 TIR data
CGM activity	7	7	Three patients lacked CGM data each visit
ITS analysis	2	2	Patient-level models only
Note: CGM = continuous glucose monitoring; HbA1c = hemoglobin A1c; ITS = interrupted time series; n = Number; TIR = time in range			

Partients' HbA1c Changes from Baseline to Visit 1 and Visit 2

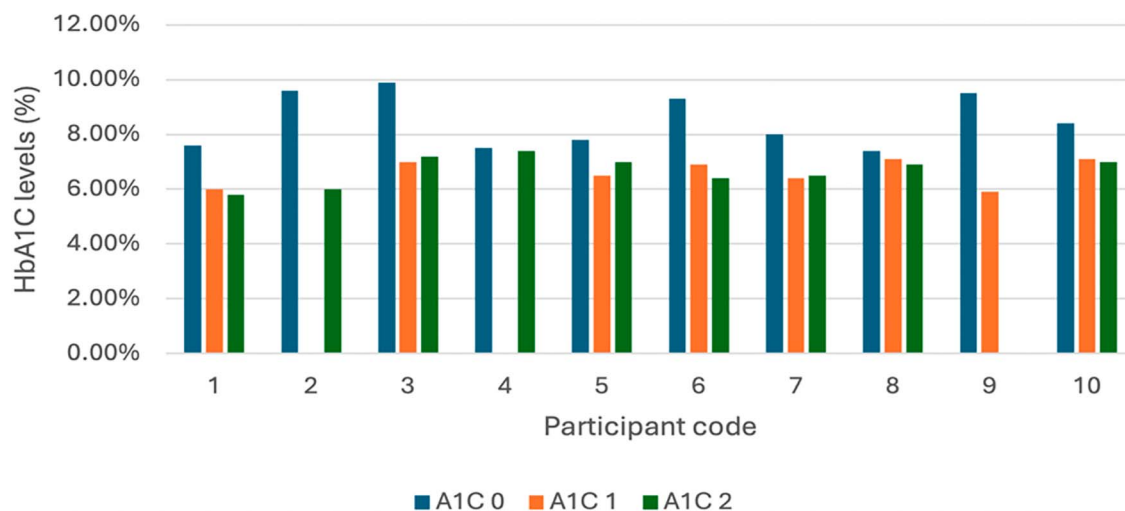


Figure 1. Bar graph of participants' HbA1c changes from baseline (visit 0) to visit 1 and visit 2. Note: HbA1c = hemoglobin A1c

Patient ID#2. The baseline HbA1c was 6.58% (95% CI: 6.48–6.86, $p < .001$). The pre-CGM trend showed a small increase ($\beta_1 = 0.0004$ per day, 95% CI: 0.0002–0.0005, $p = .0004$). The immediate post-CGM change was -0.028 (95% CI: -0.22 to 1.19 , $p = .946$). The post-CGM trend was -0.158 per day (95% CI: -0.37 to -0.10 , $p = .073$). Higher medication dosage was associated with lower HbA1c ($\beta_4 = -0.348$, 95% CI: -0.67 to -0.22 , $p = .038$). Residual autocorrelation was corrected with a corARMA ($p = 8$) structure, which provided the best fit (AIC = 54.6).

Discussion

This QI project evaluated CGM use among 10 patients with T2D treated with noninsulin antihyperglycemic medications. The intervals indicate that follow-up often exceeded the ADA-recommended follow-up 3-month visit schedule. Most participants experienced modest HbA1c reductions after CGM initiation, consistent with Ferreira et al. (2024). However, improvements were not always sustained once HbA1c reached target levels, suggesting individual variability in long-term impact. Time in range outcomes were mixed and appeared related to differences in CGM utilization, which prior studies have shown to be essential for optimizing CGM benefits (ADA, 2024b; Bergenstal et al., 2022; Moon et al., 2023).

At visit 1, all patients achieved the ADA- and AACE-recommended target of TIR $>70\%$, and most sustained it at visit 2 (ADA, 2024; AACE, n.d.). Several patients with very high baseline TIR values ($\geq 95\%$) showed only slight improvements (1–4%), suggesting limited potential for further gains when glycemic control is already near

optimal. Patient-level ITS analyses did not demonstrate statistically significant long-term changes in HbA1c trends, although the small sample limits interpretation.

Nurse practitioner and practice implications

These observations reinforce the importance of individualized diabetes management and careful assessment, which patients may be enhanced with CGM when NPs are treating patients with non-insulin-treated T2D. Our findings align with prior reports that CGM tends to be most impactful in patients with higher baseline HbA1c or suboptimal glycemia (ADA, 2024b; AACE, n.d.). Patients already meeting targets may derive less measurable benefit, underscoring the need to set realistic expectations about the benefits of CGM.

NPs can play a key role in supporting sustained engagement with CGM through structured diabetes education, integration of CGM with lifestyle counseling, and regular follow-up (Aronson et al., 2023; Cox et al., 2020; Moon et al., 2023; Price et al., 2021). Tailored support is particularly important for older adults to address barriers such as cognitive decline, dexterity issues, and technology literacy (ADA, 2024b). By providing ongoing education and monitoring, NPs can help older adults optimize CGM use within their individualized care contexts.

From a practice perspective, the project highlights the importance of monitoring CGM engagement beyond initial adoption. Although short-term improvements were common, sustained benefits likely require ongoing support. NPs are well-positioned to lead QI initiatives that identify patients at risk of disengagement and implement

Table 2. Patient-level outcomes across visits							
Patient ID	Gender	Age	HbA1c Change V1	HbA1c Change V2	TIR Change (V1 → V2)	CGM Activity V1 (%)	CGM Activity V2 (%)
1	Male	42	−1.6	−1.8	+1	78	—
2	Male	56	—	−3.6	−5	98	81
3	Female	54	−2.9	−2.7	−7	7	79
4	Male	67	—	−0.1	−1	—	—
5	Male	54	−1.30	−0.8	+4	—	55
6	Male	89	−2.4	−2.9	+9	93	93
7	Female	71	−1.6	−1.5	+2	23	29
8	Female	79	−0.3	−0.5	—	—	—
9	Male	53	−3.6	—	0	93	93
10	Nonbinary	44	−1.3	−1.4	−32	57	86

Note: HbA1c values represent change from baseline to each visit. TIR values represent change from visit 1 to visit 2. CGM activity is reported as % of time device was active. Missing values are indicated with “—.” CGM = continuous glucose monitoring; HbA1c = hemoglobin A1c; TIR = time in range

supports such as telehealth check-ins, digital reminders, or peer support groups.

Strengths

A contextual strength of this project is the sample included patients from multiple racial groups, which reflects the clinic’s patient population. However, the small sample size limits the extent to which diversity can be considered representative or allow subgroup comparisons. The use of ITS analysis to examine HbA1c trend

over 6 months provided added methodological depth. The real-world application of CGM within an endocrinology clinic offered insights into feasibility and implementation in routine practice.

Limitations

This project had both clinical and study limitations. Clinically, the absence of structured patient education, adherence monitoring, and standardized follow-up limited the ability to fully assess CGM effectiveness.

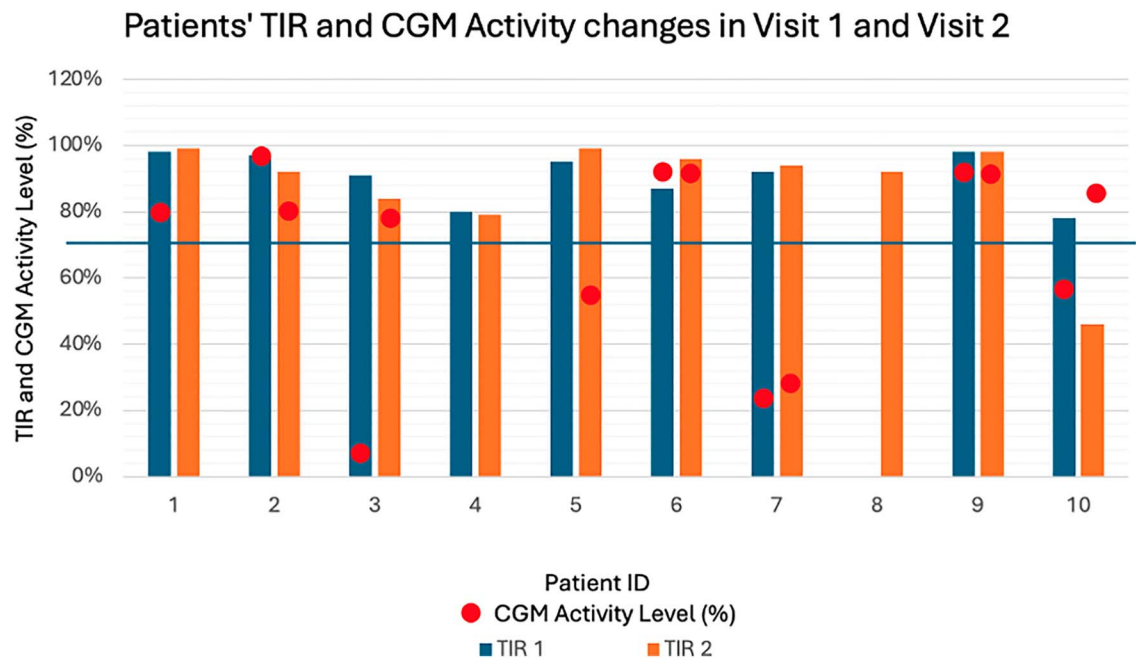


Figure 2. Bar graph of participants’ time in range and continuous glucose monitoring activity changes (%) from visit 2 to visit 1.

Variability in device use highlights the need for systematic workflows to support consistent CGM integration into routine care.

From a study perspective, the small sample size at a single endocrinology clinic limits generalizability and precluded adjustment for potential confounders such as health literacy, age, and comorbidity burden. The moderate follow-up period may also have been insufficient to capture long-term outcomes.

Finally, the ITS analyses were conducted at the patient level, illustrating individual HbA1c trajectories but not population-level effects. These exploratory models should be interpreted as hypothesis-generating, providing direction for future studies with larger sample sizes.

Implications for future practice

Future QI initiatives should further explore barriers affecting consistent CGM use, development of structured CGM training programs, and establishing routine processes for reviewing CGM data in clinical practice. Addressing these practical implementation challenges may enhance CGM's benefits, ultimately improving patient-centered diabetes management and aligning clinical practices with existing evidence-based guidelines.

Future studies should explore the long-term impact of CGM on glycemic control and its role in reducing complications associated with uncontrolled diabetes over extended follow-up periods. Investigating the economic implications of CGM implementation, including cost-effectiveness and the potential for reducing health care utilization, could inform broader health care policies and decision making regarding CGM accessibility.

Conclusion

This QI project evaluated CGM use in non-insulin-treated patients with T2D in an endocrinology clinic. HbA1c reductions were observed after CGM initiation, particularly among patients with suboptimal baseline control, but improvements were not always sustained once target HbA1c levels were reached. Variability in TIR outcomes and device engagement points to the need for structured education, adherence support, and standardized workflows to optimize CGM use in practice. NPs are well positioned to lead these efforts, given their expertise in patient education, chronic disease management, and care coordination. By delivering structured training, providing ongoing support, and integrating CGM review into routine workflows, NPs can address barriers to sustained use and tailor interventions to patient needs. Future QI initiatives should further examine NP-led strategies to enhance engagement, ensure consistent follow-up, and improve long-term CGM utilization. Larger studies are needed to confirm the long-term impact of CGM on

glycemic control, diabetes complications, and policy considerations related to access and coverage.

Authors' Contributions

N. Karimi developed the interventions, collected data, wrote the initial draft, and revised the final manuscript; D. Tolentino performed ITS analysis and revised the final manuscript; S. Choi and Y. Song revised the final manuscript; all authors provided a critical review of the final draft.

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Competing Interests

The authors report no conflicts of interest.

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